

# ClearEval: A Data-driven Benchmark for Evaluating Large Language Models in Tissue Optical Clearing

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**Abstract**—Tissue Optical Clearing (TOC) enables high-resolution 3D visualization, yet complex protocol design risks irreversible specimen damage. While LLMs excel at knowledge retrieval, their ability to synthesize viable wet-lab protocols remains under-evaluated. We introduce ClearEval, the first expert-curated, experimentally-grounded benchmark for TOC protocol design. Moving beyond semantic similarity, we evaluate 12 SOTA LLMs via a dual-track pipeline: static knowledge is assessed via 2,588 multiple-choice questions, while practical generation is evaluated on 253 open-ended scenarios through our Completeness, Correctness, and Effectiveness (CCE) framework. CCE is a text-reference-free metric that scores protocol viability through formulas derived from experimental confidence intervals rather than static texts. Results show that while models reach 80–91% accuracy in static retrieval, performance drops sharply on executable protocols (the strongest model falls from 91% knowledge to a 64-point Application Index). Failures concentrate in executable parameterization—most reliably, in matching fluorescent markers to the required labeling targets—rather than declarative recall, exposing a knowledge-execution gap in LLM-generated TOC protocols. Rather than a domain-general claim, ClearEval offers a depth-first, experimentally-grounded feasibility check—complementary to breadth-oriented protocol benchmarks—and a transferable recipe for other wet-lab subdomains. Code and data: <https://anonymous.4open.science/r/ClearEval> (de-anonymized upon acceptance).

**Index Terms**—Benchmark, Tissue optical clearing, Wet-lab protocols, Large language models

## I. INTRODUCTION

Tissue Optical Clearing (TOC) homogenizes refractive indices to enable non-destructive, sub-cellular 3D imaging via light-sheet microscopy [1], [2], preserving long-range connectivity essential for high-fidelity organ atlases [3], [4]. However, TOC protocol design is a complex multi-objective optimization problem where parameter misjudgments risk irreversible specimen damage, making traditional trial-and-error approaches both costly and inefficient.

While LLMs demonstrate proficiency in biomedical question answering [5], existing scientific benchmarks (e.g., MedQA [6], MMLU [7]) and their recent extensions [8]–[10] predominantly assess static knowledge recall without grounding in experimental validation. Furthermore, traditional semantic metrics like BLEU [11] and general-purpose evaluators [12], [13] fail to capture functional executability. Linguistically fluent protocols can remain experimentally invalid due to incorrect concentrations or processing times—subtle wet-lab errors that generic evaluators often overlook, thereby propagating hallucinations [14], [15]. Evaluating scientific LLMs

thus requires a paradigm shift: from text-based similarity to rigorous protocol validation driven by empirical evidence.

To bridge this gap, we introduce **ClearEval**, a benchmark for wet-lab grounded TOC protocol design built on an evaluation pipeline constructed from experimental data. Crucially, high knowledge scores can mask execution-level failures: a model may recall TOC facts correctly yet propose a refractive-index target or reagent combination that would ruin an irreversible specimen. A cheap pre-experimental feasibility gate is therefore valuable even when knowledge accuracy is high—e.g., GPT-5.2-Fast scores 91.1% on knowledge yet only 64.1 on executable effectiveness (Table IV). We propose the **CCE (Completeness, Correctness, Effectiveness) framework**, which validates protocol executability against empirical confidence intervals. Our main contributions are: (1) a depth-first benchmark that, within a single well-characterized subdomain (TOC), grounds protocol evaluation in experimental confidence intervals rather than text similarity; (2) a curated dataset of 2,588 domain-specific MCQs and 253 experiment-driven scenarios, with a fully documented construction pipeline (Appendix A); and (3) the data-driven CCE framework that leverages formulaic, calibrated assessment to measure LLM performance gaps in practical planning. We deliberately restrict scope to TOC to obtain physics-grounded, parameter-level ground truth that broad multi-subfield benchmarks cannot provide; external validity is discussed in Sec. VI.

## II. RELATED WORKS

**Scientific LLM evaluation.** Benchmarks for scientific LLMs have largely targeted static knowledge retrieval [5]–[7] or multidimensional reasoning [16]. Generic text metrics such as BLEU [11] and LLM-based evaluators like G-Eval [12], [13] reward linguistic fluency and can leave experimentally invalid protocols undetected, masking hallucinations [14], [15] about chemical compatibility or incubation requirements.

**Protocol-design and wet-lab benchmarks.** Breadth-oriented benchmarks such as BioProBench [17] (tens of thousands of protocols across many subfields) and LAB-Bench [18] assess protocol understanding largely via text/QA matching over many domains. ClearEval is orthogonal: it trades breadth for experiment-derived, parameter-level feasibility scoring within one subdomain, providing physics-grounded ground truth (refractive index, incubation windows, fluorophore compatibility) that QA-style benchmarks do not (Table I).

TABLE I  
POSITIONING OF CLEARVAL RELATIVE TO REPRESENTATIVE LLM  
BENCHMARKS. CLEARVAL TRADES BREADTH FOR PARAMETER-LEVEL,  
PHYSICALLY GROUNDED FEASIBILITY SCORING WITH A CALIBRATED  
JUDGE.

| Benchmark               | Task focus             | Grading signal          | Phys. ground. | Judge cal. |
|-------------------------|------------------------|-------------------------|---------------|------------|
| MMLU / MedQA            | Knowledge QA           | Exact match             | ×             | ×          |
| BioProBench             | Protocol understanding | Text / QA match         | ×             | ×          |
| LAB-Bench               | Biology research QA    | MCQ accuracy            | ×             | ×          |
| <b>ClearEval (ours)</b> | Protocol feasibility   | Parameter-level physics | ✓             | ✓          |

**LLM-as-a-judge calibration.** Automated judges exhibit documented position, verbosity, and self-preference biases [19]. We therefore restrict LLM judgment to Completeness/Correctness, ground Effectiveness in non-LLM physical checks, and report per-dimension agreement with expert consensus (Sec. V-C, Appendix B) rather than a single correlation. **Scientific agents.** Autonomous-lab systems [20], [21] aim to close the loop from hypothesis to experiment. ClearEval supplies a pre-experimental feasibility gate such agents can apply before committing irreversible reagents, grounded in the empirical parameters of Tissue Optical Clearing [1], [2], [22].

### III. THE CLEARVAL BENCHMARK

The development of ClearEval follows a structured pipeline that transforms specialized TOC domain expertise into a computational evaluation framework (Fig. 1).

#### A. Knowledge Base and Task Formulation

ClearEval formalizes wet-lab constraints into a structured knowledge system, encoded in seven curated knowledge bases covering tissue types, molecular markers, clearing-method properties, and compatibility, derived from the mainstream TOC literature (full inventory and provenance in Appendix A). To align user requirements with protocol selection, these constraints are mapped into a 6-dimensional vector space  $\mathcal{V} \in [0, 1]^6$  representing key performance indicators such as fluorescence preservation, reagent permeability, and biosafety. Within this space, 60 core scientific missions across various biological scales were defined by experts to serve as seeds for realistic task generation.

#### B. Dataset Generation Workflow

The ClearEval pipeline (Fig. 1, bottom left) instantiates expert-authored templates against the knowledge base via controlled-vocabulary substitution. For the Knowledge track, 45 expert templates (7 named-entity, 26 domain-knowledge, 12 task-scenario) are expanded into **2,588 Knowledge Tasks (MCQs)**—533 named-entity, 1,350 domain-knowledge, and 705 task-scenario—each in an 8-option format to minimize correct guessing, followed by deduplication and option shuffling. For the Application track, the 60 mission seeds are combined with the knowledge base to generate open-ended scenarios, augmented with real wet-lab challenges and quality-filtered to **253 Application Tasks (OEQs)**. The OEQs move beyond static reference texts, instead presenting constrained scenarios evaluated through an automated, experiment-based

scoring pipeline (Fig. 1, right). Templates, controlled vocabularies, and filtering criteria are documented in Appendix A.

### IV. EVALUATION FRAMEWORK

To quantify the transition from theoretical recall to functional executability, ClearEval employs a hierarchical scoring system (Table II). The Application Index ( $I_A$ ) shifts the focus from semantic similarity to **empirically grounded verification**, utilizing experimental confidence intervals to evaluate the viability of generated protocols.

TABLE II  
THE HIERARCHICAL METRIC SYSTEM OF CLEARVAL. EVALUATION  
SCALES FROM STATIC KNOWLEDGE RETRIEVAL TO  
EMPIRICALLY-GROUNDED PROTOCOL EXECUTION.

| Level                          | Dimension        | Sub-Metric                      | Method        | Max  |
|--------------------------------|------------------|---------------------------------|---------------|------|
| Knowledge ( $I_K$ )            | Static Retrieval | Accuracy                        | Deterministic | 100% |
|                                | Completeness     | Step Integrity                  | LLM-Judge     | 40%  |
| Param Detail                   |                  | LLM-Judge                       | 60%           |      |
| Application ( $I_A$ )          | Correctness      | Sequential Logic                | LLM-Judge     | 40%  |
|                                |                  | Consistency                     | LLM-Judge     | 20%  |
|                                |                  | Physics Plausibility            | LLM-Judge     | 40%  |
|                                | Effectiveness    | Method Suitability              | Vector Sim.   | 30%  |
|                                |                  | Label Compatibility             | Rule-Based    | 40%  |
|                                |                  | Trans. Validity ( $S_{trans}$ ) | Physics Eq.   | 15%  |
| Time Efficiency ( $S_{time}$ ) |                  | Physics Eq.                     | 15%           |      |
| Overall                        | Expert Score     | Harmonic Mean                   | Aggregation   | 100% |

#### A. Knowledge ( $I_K$ ) and Application ( $I_A$ ) Indices

The **Knowledge Index** ( $I_K$ ) represents the macro-averaged MCQ accuracy across domain-specific categories. To evaluate protocol design, the **Application Index** ( $I_A$ ) adopts a bottleneck formulation:

$$I_A = \min(\text{Completeness}, \text{Correctness}, \text{Effectiveness}) \quad (1)$$

This “weakest-link” approach reflects wet-lab realities, where a single critical failure in logic or chemical compatibility can render an entire experiment invalid.

#### B. Empirically-Grounded Effectiveness ( $E_{score}$ )

Unlike Completeness and Correctness, Effectiveness is scored by deterministic, non-LLM checks that read directly from the expert knowledge bases (Appendix A). From each protocol we extract its clearing method, target markers, refractive-index target, and total processing time, then compute four sub-metrics whose equal-weighted sum is  $E_{score} = S_{method} + S_{label} + S_{trans} + S_{time}$  (maxima 5/6/3/3, total 17).

**(1) Method suitability ( $S_{method}$ , max 5).** Every clearing method carries a six-dimensional capability vector  $\mathbf{v}_{ref} \in [0, 1]^6$  (fluorescence/protein preservation, dye permeability, optical clearing, geometry, operational economy, biosafety); the scenario’s requirements are parsed into a matching demand vector  $\mathbf{v}_{pred}$ .  $S_{method}$  rewards their alignment by cosine similarity, and is 0 if the proposed method is absent from the knowledge base:

$$S_{method} = 5 \cdot \max(0, \cos(\mathbf{v}_{pred}, \mathbf{v}_{ref})). \quad (2)$$

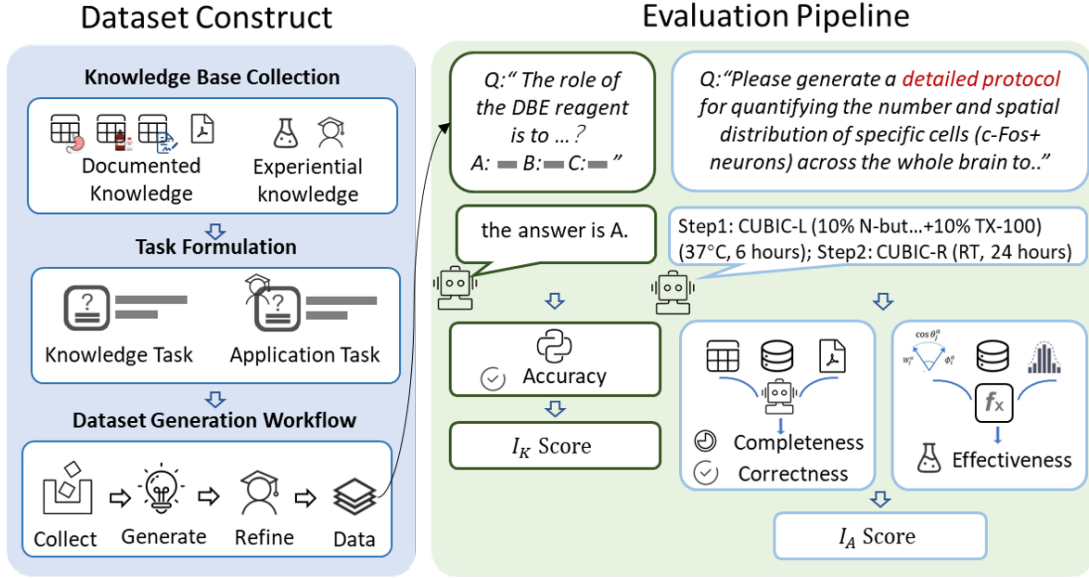


Fig. 1. Overview of the dataset construction and benchmarking pipeline. The left panel shows the workflow for formulating domain knowledge into evaluation datasets. The right panel details the evaluation process, scoring knowledge tasks on accuracy ( $I_K$ ) and application tasks on completeness, correctness, and a data-driven effectiveness metric ( $f_x$ ) grounded in experimental confidence intervals to calculate the  $I_A$  Score.

(2) **Label compatibility** ( $S_{label}$ , **max 6**). A hard constraint against unusable labeling, taken as the *minimum* of three  $[0, 6]$  rule-based checks—marker–target match, marker–fluorophore compatibility, and fluorophore–method compatibility (the last reads the quenching matrix). Any single incompatibility, e.g. a fluorophore quenched by an organic-solvent method, drives the minimum, and hence  $S_{label}$ , to 0.

(3) **Transparency validity** ( $S_{trans}$ , **max 3**). The refractive index the method achieves must match the tissue. We score the gap between the tissue’s native RI ( $RI_{tissue}$ ) and the RI the method is designed for ( $RI_{ref}$ ) with a Gaussian of width  $\sigma_{RI}$ , the per-(method, sample-scale) tolerance:

$$S_{trans} = 3 \cdot \exp\left(-\frac{(RI_{tissue} - RI_{ref})^2}{2\sigma_{RI}^2}\right). \quad (3)$$

$S_{trans}$  is gated to 0 when the method cannot physically handle the requested sample scale (no method  $\times$  tier entry in the timing base).

(4) **Time efficiency** ( $S_{time}$ , **max 3**). Let  $\Delta t$  be the deviation of the proposed total time from the literature-supported window  $[t_{min}, t_{max}]$  (with  $\Delta t=0$  inside it); a Gaussian of tolerance  $\tau$  penalizes out-of-window timing:

$$S_{time} = 3 \cdot \exp\left(-\frac{\Delta t^2}{2\tau^2}\right), \quad (4)$$

where  $\tau$  is set deterministically to  $0.2 \times$  the median clearing time. All windows, RI references, and  $\sigma_{RI}/\tau$  tolerances are literature-bounded and expert-reviewed (Appendix A). The raw  $E_{score}$  (max 17) is finally min–max normalized to  $[0, 100]$ , making it comparable with Completeness and Correctness before the  $I_A$  bottleneck.

### C. Overall Expert Score

To penalize models that memorize facts but fail under practical physical constraints, the **ClearEval Expert Score** is the harmonic mean:  $\text{Score} = 2I_K I_A / (I_K + I_A)$ . Given the bottleneck formulation of  $I_A$ , a high score demands both broad theoretical recall and strict experimental compliance.

## V. EXPERIMENTS

### A. Models and Settings

We benchmark 12 LLMs across the GPT, Gemini, Claude, DeepSeek/GLM, and Qwen families. For all evaluated API models we set temperature to 0.0 for both Knowledge and Application tasks to maximize reproducibility; reasoning variants additionally run in their native thinking mode (Table III). Exact provider snapshot strings, context windows, maximum output tokens, and API call dates are released in the repository configuration (Appendix C).

### B. Main Results: The Knowledge-Execution Gap

Table IV reveals a stark capability imbalance between static recall and wet-lab execution: Knowledge indices cluster at 80–91, yet the bottleneck Application Index  $I_A$  ranges from 64 down to 18. This is visually highlighted in Fig. 2: the left panel shows all models with uniformly high, tightly clustered Knowledge proficiency, whereas Application Performance (right) is highly erratic. Models generally maintain high procedural completeness (79 on average) but collapse on Correctness (50) and Effectiveness (59); for the three strongest models the binding constraint is Effectiveness, for the other nine it is Correctness—though that dimension is partly deflated by the weakly-calibrated *co\_chem* sub-score (Sec. V-C). The most reliable single failure is label suitability ( $S_{label}$ ): models

TABLE III

DECODING CONFIGURATION FOR THE 12 EVALUATED MODELS. “REASONING” DENOTES WHETHER THE RUN USED THE MODEL’S NATIVE THINKING MODE. TEMPERATURE IS 0.0 THROUGHOUT; FULL PER-MODEL SNAPSHOTS AND CONTEXT WINDOWS ARE IN THE RELEASED CONFIGURATION (APPENDIX C).

| Model               | Provider  | Reasoning | Temp. |
|---------------------|-----------|-----------|-------|
| GPT-5.2-Fast        | OpenAI    | –         | 0.0   |
| GPT-5.2-Think       | OpenAI    | ✓         | 0.0   |
| Gemini-3-Flash      | Google    | –         | 0.0   |
| Gemini-3-Pro        | Google    | ✓         | 0.0   |
| Claude-4.6-Sonnet   | Anthropic | –         | 0.0   |
| GLM-4.7             | Zhipu     | –         | 0.0   |
| DeepSeek-V3.2       | DeepSeek  | –         | 0.0   |
| DeepSeek-V3.2-Think | DeepSeek  | ✓         | 0.0   |
| Qwen3-max           | Alibaba   | –         | 0.0   |
| Qwen3-235B-A22B     | Alibaba   | –         | 0.0   |
| Qwen3-32B           | Alibaba   | –         | 0.0   |
| Qwen3-14B           | Alibaba   | –         | 0.0   |

propose fluorescent markers that miss the scenario’s required labeling targets (Sec. V-D), exposing a severe knowledge–execution gap.

### C. Calibration of the Automated Metrics

We validate the automated CCE scores against blinded expert consensus on 240 protocols (20 scenarios  $\times$  12 models). Because Pearson correlation alone ignores absolute bias, we report, per dimension, Pearson  $r$ , Spearman  $\rho$ , ICC(2,1), quadratic-weighted  $\kappa$ , and Krippendorff’s  $\alpha$  (Table V; full 14-row sub-dimension breakdown in Appendix B). The aggregate CCE dimensions align well with experts (ICC 0.75/0.65/0.81 for Completeness/Correctness/Effectiveness; Overall  $r = 0.84$ ). Crucially, agreement is *uneven* across the Effectiveness sub-metrics: the empirically-grounded label-compatibility ( $S_{label}$ , ICC 0.86) and timing ( $S_{time}$ , ICC 0.82) checks align strongly with experts, and transparency ( $S_{trans}$ , ICC 0.58) moderately, whereas the cosine-similarity method-suitability score ( $S_{method}$ , ICC 0.04,  $r = 0.11$ ) is near-degenerate—it is therefore reported as a soft preference signal rather than a calibrated score. Among the two LLM-judged dimensions, the weakest agreement is on chemical correctness ( $co\_chem$ , ICC 0.32). We thus restrict LLM-as-judge to Completeness and Correctness—noting that  $co\_chem$  is the least reliable sub-score and reading the corresponding failure as suggestive (Sec. V-D)—and rely on the non-LLM physical checks for Effectiveness, where the dominant failure ( $S_{label}$ ) is also the best-calibrated (ICC 0.86).

### D. Fine-Grained Diagnosis: Where Execution Fails

Averaging normalized sub-scores over all 253 scenarios and 12 models, the *reliably*-measured deficit concentrates in label suitability (Fig. 4). **Label suitability** ( $S_{label}$ , mean 0.23) is a hard minimum over a marker–target match and two fluorophore-compatibility checks, and 66% of protocols score 0. Decomposing these zeros from the scorer’s own breakdown shows the binding cause is **marker–target mismatch**—

the fluorescent markers models propose do not match the scenario’s required labeling targets—rather than fluorophore quenching, which is the minority case. Because  $S_{label}$  is our best-calibrated Effectiveness check (ICC 0.86, Sec. V-C), this is the most trustworthy execution failure we observe. The nominally lower chemical-safety score ( $co\_chem$ , mean 0.07) requires care: inspecting the judge’s rationales shows it is dominated by penalties for the *absence* of safety/handling caveats (fume-hood use, solvent-compatible containers) that our generation prompt never requested; given this and its weak calibration (ICC 0.32), we treat  $co\_chem$  as suggestive rather than a capability measure. By contrast, transparency / refractive-index matching ( $S_{trans}$ , 0.64) and timing ( $S_{time}$ , 0.59) are only moderately impaired—though  $S_{time}$  is sensitive to the width of the literature time windows, and individual estimates can still be wildly off (occasionally exceeding 800 hours against a 0–200 hour window). The cosine method-suitability score ( $S_{method}$ , 0.89) appears high but, being near-degenerate against experts (ICC 0.04), is discounted. In short, the most reliable wet-lab constraint these LLMs miss is matching fluorescent markers to the scenario’s required targets.

### E. Inference-Time Robustness Protocol

A natural follow-up question is whether the knowledge–execution gap narrows once standard inference-time techniques are applied, rather than being an artifact of greedy decoding. To make this directly testable on ClearEval, we release a turnkey harness that scores four recipes against the greedy default on the full 253-scenario Application set: (i) **few-shot** (3 in-context exemplars); (ii) **self-consistency** ( $k=5$  samples at temperature 0.7, aggregated by method-majority then continuous-field median); (iii) **KB-RAG**, prepending retrieved knowledge-base context (method properties and compatibility records, with numeric RI/time targets withheld so the answer key is never leaked); and (iv) **KB-RAG + self-consistency**. Because every recipe is scored by the same calibrated CCE pipeline (Sec. V-C), the bottleneck Application Index  $I_A$ —reported with 95% confidence intervals over samples—is directly comparable across recipes and against Table IV. We treat KB-RAG as an upper bound on grounding benefit and leave full literature-RAG over an external corpus to future work. The harness and the exact run protocol are released (Appendix C); a complete multi-model sweep under paid inference is ongoing and will be reported in an extended version. We deliberately do not report partial numbers here to avoid drawing conclusions from an incomplete sweep.

## VI. DISCUSSION AND LIMITATIONS

Within TOC, ClearEval shows the bottleneck is not declarative retrieval but executable parameterization—most reliably, matching fluorescent markers to the required labeling targets, with refractive-index and timing errors secondary. Whether this pattern generalizes across wet-lab subdomains is an open question our single-subdomain design cannot settle.

**External validity.** ClearEval covers one subdomain (TOC). Our claims are therefore about TOC protocol feasibility,

TABLE IV  
PERFORMANCE OF STATE-OF-THE-ART MODELS ON CLEARVAL. EVALUATION COVERS KNOWLEDGE (MCQ) AND APPLICATION (OEQ).  $I_K$  REPRESENTS THE AVERAGE OF THE KNOWLEDGE SUBSET,  $I_A$  REPRESENTS THE MINIMUM SCORE ACROSS THE APPLICATION SUBSET, AND TOTAL IS THEIR HARMONIC MEAN.

| Models                    | Knowledge (MCQ) |             |             |             | Application (OEQ) |             |             |             | Total       |
|---------------------------|-----------------|-------------|-------------|-------------|-------------------|-------------|-------------|-------------|-------------|
|                           | Tissue          | Reagent     | Method      | $I_K$       | Com.              | Cor.        | Eff.        | $I_A$       |             |
| <i>Proprietary Models</i> |                 |             |             |             |                   |             |             |             |             |
| GPT-5.2-Fast              | <b>91.1</b>     | <b>88.7</b> | 93.5        | <b>91.1</b> | <u>96.8</u>       | <b>67.9</b> | <b>64.1</b> | <b>64.1</b> | <b>75.3</b> |
| GPT-5.2-Think             | <u>90.5</u>     | <u>87.3</u> | <u>93.6</u> | <u>90.5</u> | 96.6              | <u>67.6</u> | 63.5        | <u>63.5</u> | <u>74.6</u> |
| Gemini-3-Flash            | <u>78.7</u>     | 81.8        | 91.1        | 83.8        | 75.1              | 55.8        | 63.4        | 55.8        | 67.0        |
| Gemini-3-Pro              | 88.8            | 77.2        | 86.7        | 84.2        | 81.2              | 60.5        | <u>63.8</u> | 60.5        | 70.4        |
| Claude-4.6-Sonnet         | 87.6            | 82.9        | <b>94.9</b> | 88.5        | <b>98.3</b>       | 63.9        | <u>63.2</u> | 63.2        | 73.7        |
| <i>Open-Weight Models</i> |                 |             |             |             |                   |             |             |             |             |
| GLM-4.7                   | 79.3            | 80.2        | 93.5        | 84.3        | 74.1              | 55.2        | 62.3        | 55.2        | 66.7        |
| DeepSeek-V3.2             | 79.9            | 77.9        | 91.7        | 83.1        | 75.8              | 46.4        | 50.0        | 46.4        | 59.6        |
| DeepSeek-V3.2-Think       | 85.3            | 83.8        | 93.5        | 87.5        | 75.3              | 46.0        | 49.2        | 46.0        | 60.3        |
| Qwen3-max                 | 79.9            | <u>87.3</u> | 93.1        | 86.8        | 84.3              | 54.7        | 62.6        | 54.7        | 67.1        |
| Qwen3-235B-A22B           | 77.5            | 81.2        | 89.1        | 82.6        | 74.8              | 33.8        | 56.0        | 33.8        | 47.9        |
| Qwen3-32B                 | 73.4            | 78.6        | 89.9        | 80.6        | 59.6              | 30.2        | 52.4        | 30.2        | 44.0        |
| Qwen3-14B                 | 73.4            | 77.8        | 88.4        | 79.9        | 56.3              | 18.4        | 55.3        | 18.4        | 29.9        |

TABLE V  
AGREEMENT OF AUTOMATED VS. BLINDED EXPERT-CONSENSUS SCORES ON 240 PROTOCOLS (20 SCENARIOS  $\times$  12 MODELS). ICC=ICC(2,1); QWK=QUADRATIC-WEIGHTED  $\kappa$ ;  $\alpha$ =INTERVAL KRIPPENDORFF. FULL 14-ROW BREAKDOWN IN APPENDIX B.

| Dimension                        | Pearson $r$ | Spearman | ICC  | QWK  | $\alpha$ |
|----------------------------------|-------------|----------|------|------|----------|
| Completeness                     | 0.77        | 0.79     | 0.75 | 0.70 | 0.75     |
| Correctness                      | 0.68        | 0.66     | 0.65 | 0.65 | 0.64     |
| Effectiveness                    | 0.83        | 0.81     | 0.81 | 0.80 | 0.80     |
| Overall                          | 0.84        | 0.84     | 0.79 | 0.79 | 0.78     |
| <i>Effectiveness sub-metrics</i> |             |          |      |      |          |
| $S_{method}$ (cosine)            | 0.11        | 0.15     | 0.04 | 0.10 | -0.04    |
| $S_{label}$ (rule)               | 0.86        | 0.82     | 0.86 | 0.85 | 0.86     |
| $S_{trans}$ (physics)            | 0.63        | 0.46     | 0.58 | 0.52 | 0.56     |
| $S_{time}$ (physics)             | 0.84        | 0.77     | 0.82 | 0.80 | 0.82     |

not LLM scientific competence in general. Generalizing the methodology requires re-curating subdomain-specific physical knowledge bases (refractive indices, incubation windows, compatibility matrices), which is non-trivial (Appendix A); we offer CCE as a transferable recipe rather than a finished multi-domain benchmark.

**Complementarity.** Depth-first ClearEval and breadth-first benchmarks such as BioProBench [17] are complementary: the former contributes experiment-derived, parameter-level feasibility scoring within a subdomain, the latter broad coverage. Together they give a fuller picture of LLM protocol-design ability than either alone.

**Toward grounded agents.** For LLMs to become viable in specialized wet-lab domains, they must internalize experimen-

tal constraints—such as reagent diffusion scales—within their reasoning loops; ClearEval can serve as a pre-experimental feasibility gate for such agents before irreversible reagent commitment.

**Limitations.** ClearEval remains a static paradigm. Real-world wet labs involve stochastic variables (reagent degradation, tissue heterogeneity) that are hard to model in silico; our expert grounding is literature-bounded and refined by expert review rather than exhaustively re-measured; and the cosine method-suitability sub-score is weakly calibrated (Sec. V-C). Future work should integrate dynamic hardware feedback so agents iteratively optimize protocols from real-time signals.

## VII. CONCLUSION

We presented ClearEval, an expert-curated benchmark for TOC protocol design that shifts evaluation from semantic similarity to empirically grounded functional validation. By leveraging a dual-track pipeline and a data-driven CCE framework, ClearEval exposes a significant knowledge-execution gap in current LLMs on TOC protocol design. Our results indicate that while models excel at knowledge recall, they struggle with the marker-target matching and physical-feasibility constraints required for wet-lab execution. We anticipate that ClearEval—and the transferable CCE recipe—will facilitate the development of reliable scientific agents, starting with tissue-clearing workflows.

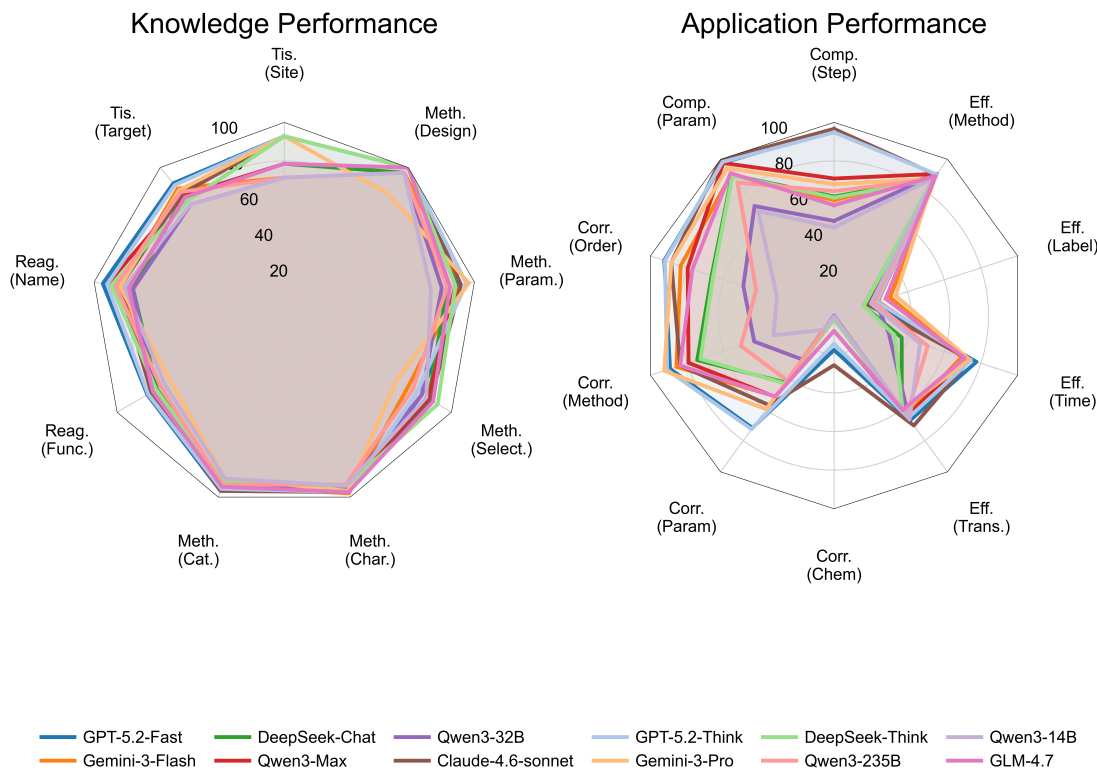


Fig. 2. Performance comparison of 12 LLMs on the ClearEval benchmark. The left panel shows Knowledge Performance ( $I_K$ ) across domain-specific attributes (Tissue, Reagent, Method). The right panel details Application Performance ( $I_A$ ), evaluated on completeness, correctness, and an Empirically-Grounded effectiveness metric.

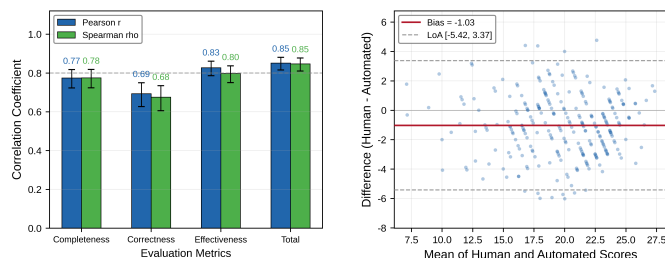


Fig. 3. Calibration of automated metrics against expert consensus ( $n=240$ ). (a) Per-dimension correlation with expert ratings. (b) Bland-Altman plot showing high absolute agreement.

## APPENDIX A

### KNOWLEDGE-BASE CURATION AND EXPERT GROUNDING

**Expert panel and consensus.** The knowledge bases and the CCE rubric were curated by a panel of tissue-clearing wet-lab experts together with a knowledge-graph specialist. Application protocols were first scored blind against the rubric and then reconciled by panel consensus; the released scores are per-protocol consensus values.

**Knowledge bases.** Grounding is encoded in seven JSON knowledge bases: reference refractive index for 18 clearing methods (CUBIC, ScaleS, MACS, iDISCO+, FDISCO, SOLID, SeeDB2, PEGASOS, BoneClear, Ce3D, TDE, ClearT2, ClearSee, EyeCi, FOCM, SWITCH, uDISCO, Fly-

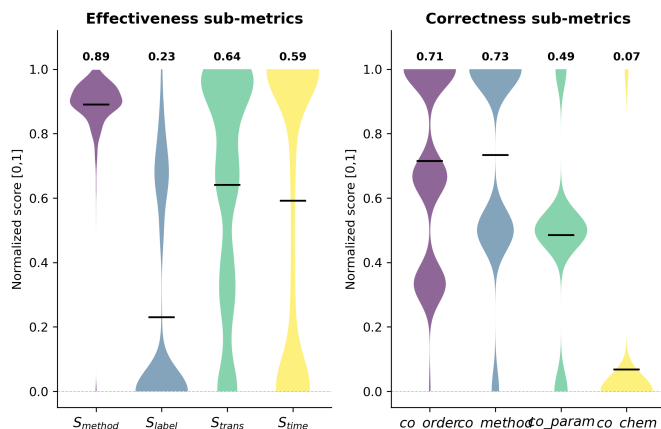


Fig. 4. Sub-metric score distributions over all 253 scenarios, pooled across the 12 models and normalized to  $[0, 1]$  (means annotated). Left: Effectiveness sub-metrics—label suitability ( $S_{label}$ , 0.23) is the dominant *reliable* failure, while cosine method-suitability ( $S_{method}$ , 0.89) is high but weakly calibrated (ICC 0.04). Right: Correctness sub-metrics—the chemical-safety score ( $co_{chem}$ , 0.07) collapses, but chiefly because the judge penalizes missing safety caveats (Sec. V-D), not chemistry errors.

Clear); per-method RI tolerance  $\sigma_{RI}$ ; per-method incubation-time tolerance  $\tau$ ; a timing base of 84 (method, sample-tier) rows giving the viable window  $[t_{min}, t_{max}]$ ; a marker $\times$ method fluorophore-compatibility matrix; a tissue table; and tissue-RI references.

**Provenance.** Each timing row records a literature source key, source URL, evidence locator, confidence level, and review status, giving an audit trail from each parameter to its literature basis (Table VI). RI references cite their originating protocol (e.g., CUBIC RI=1.46, Susaki et al., *Nat. Protoc.*). Timing tolerances are derived deterministically as  $\tau=0.2\times$  the median clearing time. Parameters are literature-bounded and refined by expert review rather than exhaustively re-measured.

**Operationalization.** The Effectiveness terms read directly from these bases:  $S_{method}$  from tissue/method preference vectors (cosine);  $S_{label}$  from the fluorophore-compatibility matrix (hard zero on quenching);  $S_{trans}$  from RI references and  $\sigma_{RI}$ , gated to zero when a (method, tier) pair is unsupported; and  $S_{time}$  from  $[t_{min}, t_{max}]$  and  $\tau$ .

TABLE VI  
SCHEMA OF THE TIMING KNOWLEDGE BASE (84 ROWS). EVERY ROW IS TRACEABLE TO A LITERATURE SOURCE AND A REVIEW DECISION.

| Field                         | Role / example                  |
|-------------------------------|---------------------------------|
| method, tier                  | e.g. CUBIC, whole-organ tier    |
| $[t_{min}, t_{med}, t_{max}]$ | viable incubation window (h)    |
| source_key, source_url        | e.g. CUBIC_NATPROT_2015         |
| evidence_location             | locator within the cited source |
| confidence                    | high / medium / low             |
| review_status                 | added / modified / removed      |

## APPENDIX B JUDGE CALIBRATION DETAILS

Table VII reports the full per-sub-dimension agreement between automated and blinded expert-consensus scores ( $n=240$ ). The two LLM-judged dimensions use a fixed rubric (eval\_oeq\_teacher\_rubric.txt, eval\_completeness\_checklist.txt, released in the repository) at temperature 0.0. Because raw per-annotator sheets were not preserved, we report expert-consensus-vs-automated agreement rather than human-human inter-annotator reliability.

## APPENDIX C REPRODUCIBILITY AND ARTIFACT MANIFEST

**Artifacts.** The repository releases the 253 Application scenarios and 2,588 MCQs; the seven knowledge bases; all 12 model generations and their evaluation results; the generation and judge prompts; and the scoring/plotting code, with an English README, a requirements.txt, and UTF-8 knowledge-base fields.

**Determinism.** All evaluated API models run at temperature 0.0 (Table III); because hosted models are non-deterministic and snapshot aliases drift, we release the raw model responses so results are replayable even when regeneration is not bit-identical. Provider snapshot strings and API-call dates are recorded in the released configuration.

**Inference-time run protocol (Sec. V-E).** Few-shot uses 3 exemplars; self-consistency draws  $k=5$  samples at temperature 0.7 aggregated by method-majority then continuous-field

TABLE VII  
FULL PER-DIMENSION AGREEMENT OF AUTOMATED VS. EXPERT-CONSENSUS SCORES ( $n=240$ ). ICC=ICC(2,1); QWK=QUADRATIC-WEIGHTED  $\kappa$ ;  $\alpha$ =INTERVAL KRIPPENDORFF.

| Dimension            | $r$  | $\rho$ | ICC  | QWK  | $\alpha$ |
|----------------------|------|--------|------|------|----------|
| $c_{step}$           | 0.71 | 0.72   | 0.70 | 0.54 | 0.70     |
| $c_{param}$          | 0.71 | 0.71   | 0.68 | 0.60 | 0.67     |
| <b>Completeness</b>  | 0.77 | 0.79   | 0.75 | 0.70 | 0.75     |
| $co_{order}$         | 0.66 | 0.65   | 0.61 | 0.58 | 0.60     |
| $co_{method}$        | 0.46 | 0.44   | 0.40 | 0.39 | 0.37     |
| $co_{param}$         | 0.55 | 0.56   | 0.53 | 0.48 | 0.53     |
| $co_{chem}$          | 0.41 | 0.32   | 0.32 | 0.37 | 0.25     |
| <b>Correctness</b>   | 0.68 | 0.66   | 0.65 | 0.65 | 0.64     |
| $S_{method}$         | 0.11 | 0.15   | 0.04 | 0.10 | -0.04    |
| $S_{label}$          | 0.86 | 0.82   | 0.86 | 0.85 | 0.86     |
| $S_{trans}$          | 0.63 | 0.46   | 0.58 | 0.52 | 0.56     |
| $S_{time}$           | 0.84 | 0.77   | 0.82 | 0.80 | 0.82     |
| <b>Effectiveness</b> | 0.83 | 0.81   | 0.81 | 0.80 | 0.80     |
| <b>Overall</b>       | 0.84 | 0.84   | 0.79 | 0.79 | 0.78     |

median; KB-RAG prepends retrieved compatibility/method records with numeric RI/time targets withheld; reported  $I_A$  carries 95% CIs over samples.

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